

Chapter: Life Processes

1. The kidneys in human beings are a part of the system for
 - a. Nutrition.
 - b. Excretion.
 - c. Respiration.
 - d. transportation.
2. The xylem in plants are responsible for
 - a. transport of water.
 - b. transport of amino acids.
 - c. transport of food.
 - d. transport of oxygen.
3. The autotrophic mode of nutrition requires
 - a. carbon dioxide and water.
 - b. sunlight.
 - c. Chlorophyll.
 - d. all of the above.
4. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in
 - a. cytoplasm.
 - b. Chloroplast.
 - c. Mitochondria.
 - d. nucleus.
5. How are fats digested in our bodies? Where does this process take place?
6. What is the role of saliva in the digestion of food?
7. What are the necessary conditions for autotrophic nutrition and what are its byproducts?
8. What are the differences between aerobic and anaerobic respiration? Name some organisms that use the anaerobic mode of respiration.
9. How are the alveoli designed to maximise the exchange of gases?
10. What would be the consequences of a deficiency of haemoglobin in our bodies?
11. Describe double circulation of blood in human beings. Why is it necessary?
12. What are the differences between the transport of materials in xylem and phloem?
13. Compare the functioning of alveoli in the lungs and nephrons in the kidneys with respect to their structure and functioning.

Chapter: How do Organisms Reproduce?

1. Asexual reproduction takes place through budding in
 - a. Amoeba
 - b. Yeast
 - c. Plasmodium
 - d. Leishmania.
2. Which of the following is not a part of the female reproductive system in human beings?
 - a. Ovary
 - b. Uterus
 - c. Vas deferens
 - d. Fallopian tube
3. The anther contains
 - a. Sepals
 - b. Ovules
 - c. Pistil
 - d. pollen grains
4. What are the advantages of sexual reproduction over asexual reproduction?
5. What are the functions performed by the testis in human beings?
6. Why does menstruation occur?
7. Draw a labelled diagram of the longitudinal section of a flower.
8. What are the different methods of contraception?
9. How are the modes for reproduction different in unicellular and multicellular organisms?
10. How does reproduction help in providing stability to populations of species?
11. What could be the reasons for adopting contraceptive methods?

Chapter: Heredity

1. A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers, but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as
 - a. TTWW
 - b. TTww
 - c. TtWW
 - d. TtWw
2. A study found that children with light-coloured eyes are likely to have parents with light-coloured eyes. On this basis, can we say anything about whether the light eye colour trait is dominant or recessive? Why or why not?
3. Outline a project which aims to find the dominant coat colour in dogs.
4. How is the equal genetic contribution of male and female parents ensured in the progeny?